PHYSICS

Paper 1 (Theory)

Maximum Marks: 70 **Time Allowed: Three Hours**

Reading Time: Additional Fifteen Minutes

Instructions to Candidates

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during reading time.
3. This question paper has 10 printed pages.
4. It has twenty questions in this paper. Answer all questions.
5. There are four sections in the paper: A, B, C and D. Internal choices have been provided in two questions each in Sections B, C and D.
6. Section A consists of one question having fourteen subparts of one mark each.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B consists of seven questions of two marks each.
9. Section C consists of nine questions of three marks each.
10. Section D consists of three questions of five marks each.
11. The intended marks for questions are given in brackets [ ].
12. A list of useful constants and relations is given below these instructions.
13. A simple scientific calculator without a programmable memory may be used for calculations.

USEFUL CONSTANTS AND RELATIONS

1.	Planck's constant	h	=	$6{\cdot}6 \times 10^{-34}$ J s
2.	Speed of light in vacuum	c	=	3×10^{8} m s^{-1}
3.	Charge of a proton	e	=	$1{\cdot}6 \times 10^{-19}$ C
4.	Mass of an electron	m_e	=	$9{\cdot}1 \times 10^{-31}$ kg
5.	Permittivity of vacuum	ε_0	=	$8{\cdot}85 \times 10^{-12}$ F m^{-1}
6.	Permeability of vacuum	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
7.	One atomic mass unit	1 u	=	$1{\cdot}66 \times 10^{-27}$ kg = 931·5 MeV/c^2
8.	One electron volt	1 eV	=	$1{\cdot}6 \times 10^{-19}$ J
9.		1 MeV	=	$1{\cdot}6 \times 10^{-13}$ J
10.	Rydberg constant	R	=	$1{\cdot}097 \times 10^{7}$ m^{-1}

Note: The questions in this sample paper follow the format of the ISC Specimen Question Paper 2027. The weightage allocated to each unit in the syllabus has been strictly adhered to.

SECTION A – 14 MARKS

Question 1

(A) In questions (i) to (vii), choose the correct alternative (a), (b), (c) or (d) for each of the questions given below.

(i) Point charges of +30 nC, −10 nC and +5 nC lie inside a large closed spherical surface, and a further charge of +40 nC lies outside it. The net electric flux through the surface is: *(Apply)* [1]

(a) $5{\cdot}1 \times 10^3$ N m^2 C^{-1}

(b) $2{\cdot}8 \times 10^3$ N m^2 C^{-1}

(c) $7{\cdot}3 \times 10^3$ N m^2 C^{-1}

(d) zero

(ii) Given below are two statements marked, Assertion and Reason. Read the two statements and choose the correct option. [1]

Assertion: The kinetic energy of a charged particle describing a circular path in a uniform magnetic field changes continuously with time.

Reason: The velocity of such a particle changes continuously with time, because its direction keeps changing. *(Analyse)*

(a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Assertion is false and Reason is true.

(iii) A 50 Hz a.c. voltage source is applied to a series combination of a resistor of resistance P and a component B of reactance Y. The current leads the voltage by $\pi/4$. Which of the following is true? *(Apply)* [1]

(a) B is an inductor and $Y = P$

(b) B is a capacitor and $Y = 2P$

(c) B is a capacitor and $Y = P$

(d) B is an inductor and $Y = 2P$

(iv) Identify the correct statements about a photon. *(Understand)* [1]

(P) Its rest mass is zero and its energy is entirely kinetic.

(Q) Its energy is $h\nu$ and its momentum is h/λ.

(R) Its speed in vacuum depends on its frequency.

(S) It is deflected by an electric field.

(a) Both (Q) and (R)

(b) Both (P) and (Q)

(c) Both (R) and (S)

(d) Both (P) and (S)

(v) Select the odd one out from the following dopants. *(Analyse)* [1]

(a) Boron

(b) Indium

(c) Gallium

(d) Arsenic

(vi) Given below are two statements marked, Assertion and Reason. Read the two statements and choose the correct option. [1]

Assertion: The graph of *v* against *u* for the real images formed by a concave mirror lies in the third quadrant.

Reason: For a real image formed by a concave mirror, both *u* and *v* are negative in the Cartesian sign convention. *(Analyse)*

(a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Assertion is false and Reason is true.

(vii) Monochromatic light is used in Young's double slit experiment and then in a single-slit diffraction experiment. Identify the correct statements. *(Understand)* [1]

(P) The bright fringes of the double-slit pattern are of equal width.

(Q) In the single-slit pattern the central maximum is the widest and the brightest.

(R) All the maxima of the single-slit pattern are of equal intensity.

(S) The central fringe of the double-slit pattern is twice as wide as the others.

(a) Both (Q) and (R)

(b) Both (P) and (S)

(c) Both (R) and (S)

(d) Both (P) and (Q)

(B) Answer the following questions briefly:

(i) A parallel plate capacitor is charged to a potential *V* by a battery. The battery is then disconnected and the plates are pulled slightly farther apart. Does the capacitance increase, decrease or remain the same? Explain. *(Understand)* [1]

(ii) A 24 V battery, a 6 Ω resistor and an ammeter are connected in series. The ammeter has a coil of resistance 180 Ω shunted by 20 Ω. Find the reading of the ammeter. *(Apply)* [1]

(iii) A single slit of width 0·10 mm gives a certain diffraction pattern with light of wavelength 400 nm. To what width must the slit be changed to obtain the same pattern with light of wavelength 600 nm? *(Apply)* [1]

(iv) How does the refractive index of the material of a prism depend on the angle of the prism and the angle of minimum deviation? *(Recall)* [1]

(v) An electron revolves in the second orbit of a hydrogen atom. Given that the radius of the first orbit is $0{\cdot}53 \times 10^{-10}$ m, calculate the de Broglie wavelength of the electron. *(Apply)* [1]

(vi) State two conditions necessary for nuclear fusion. *(Recall)* [1]

(vii) Name the diode that is used as a voltage regulator and state the bias in which it is operated. *(Recall)* [1]

SECTION B – 14 MARKS

Question 2

(i) The electric potential V varies with distance x as shown in Figure 1. Construct a graph of the electric field strength E as a function of x. *(Analyse)* [2]

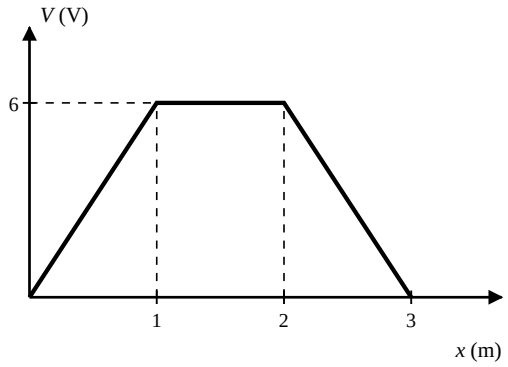


Figure 1

OR

(ii) Figure 2 shows a closed surface S and four point charges. Calculate the electric flux emanating from the closed surface S. *(Apply)* [2]

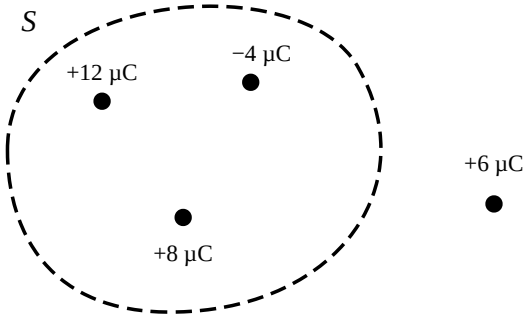


Figure 2

Question 3

Three identical capacitors, each of capacitance 6 µF, are connected in parallel across a 12 V battery. (a) Calculate the energy stored in the system. (b) Find the ratio C_s/C_p of the equivalent capacitance when the three are connected in series to that when they are connected in parallel. *(Apply)* [2]

Question 4

(i) A heater takes 20 minutes to boil a given amount of water. Its coil is cut and one-fourth of its length is used in the heater. How much time will it now take to boil the same amount of water using the same source, starting from the same initial temperature? *(Apply)* [2]

OR

(ii) (a) Name a material that has a very low temperature coefficient of resistance. (b) Give an example of a non-Ohmic conductor and state how its *V–I* graph differs from that of an Ohmic one. *(Recall)* [2]

Question 5

(i) A solenoid of 40 turns per cm carries a current of 3 A. Find the magnetic field intensity at its centre. *(Apply)* [1]

(ii) Draw a labelled graph showing how the inductive reactance X_L and the capacitive reactance X_C vary with the frequency of the alternating voltage source. *(Understand)* [1]

Question 6

Young's double-slit experiment is performed first in air and then in a liquid. The 6th bright fringe in the liquid is found to coincide with the 4th dark fringe in air. Find the refractive index of the liquid. *(Apply)* [2]

Question 7

Analyse the following two situations and name the electromagnetic wave responsible in each case. *(Analyse)*

(i) The glass of a greenhouse lets the sunlight in but does not let the heat radiated by the warm soil and plants escape. Which radiation is trapped? *(Analyse)* [1]

(ii) Aircraft navigation depends on radar. Name the electromagnetic wave used in radar and write its frequency range. *(Analyse)* [1]

Question 8

A photoelectric apparatus with a metal of unknown work function is irradiated with light of wavelength 250 nm. The photocurrent falls to zero when the collector plate is at −1·2 V. Determine the work function of the metal. *(Apply)* [2]

SECTION C – 27 MARKS

Question 9

(i) Show that the electric field at a point on the axial line of a short electric dipole, at a distance r from its centre, is given by $E = \frac{1}{4\pi\varepsilon_0} \frac{2pr}{(r^2 - l^2)^2}$, where the symbols have their usual meaning. *(Understand)* [3]

OR

(ii) A student is solving a problem on Gauss's law. The examiner gives instructions and the student responds. State whether each response is correct or incorrect and give a reason. [3]

(a) **Examiner:** Draw a Gaussian surface to find the field at a point due to a uniformly charged infinite plane sheet.

Student: Draws a cylinder piercing the sheet, with its end caps parallel to the sheet and equidistant from it.

(b) **Examiner:** Suggest a way to double the flux through a given closed surface.

Student: Doubles the area of the surface.

(c) **Examiner:** Find the flux through a square plate of side 10 cm due to a charge of 2 μC placed 5 cm directly above its centre.

Student: The charge is not enclosed by the plate, so the flux is zero. *(Evaluate)*

Question 10

(i) Using Kirchhoff's laws, calculate the current in each branch of the network of Figure 3. (The resistance shown in each outer branch includes the internal resistance of its cell, which is given beside the cell.) *(Apply)* [3]

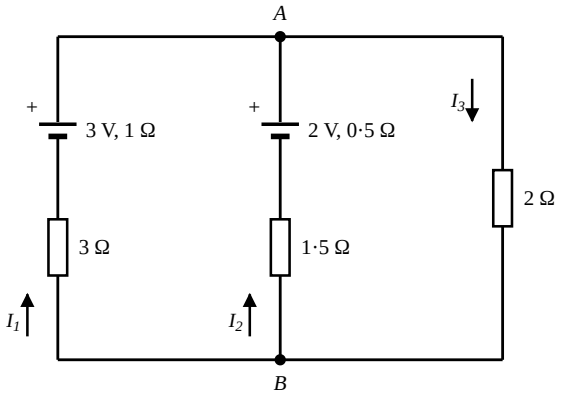


Figure 3

OR

(ii) In the metre bridge of Figure 4, the resistance in the left gap is 2 Ω and the null point is found at 40 cm from the left end. What resistance should be connected to the unknown resistance *X*, and how, so that the null point shifts to 50 cm from the left end? *(Analyse)* [3]

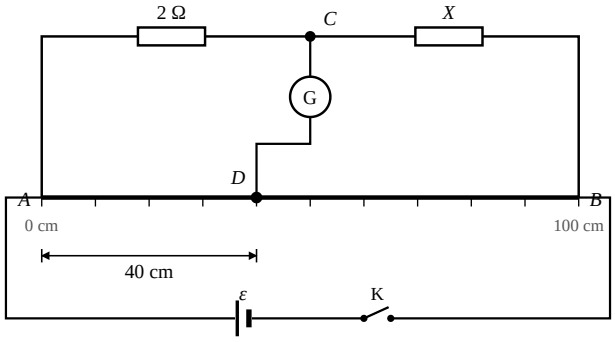


Figure 4

Question 11

Using the Biot–Savart law, obtain an expression for the magnetic field at the centre of a circular coil of radius *R* having *N* turns and carrying a current *I*. *(Understand)* [3]

Question 12

(i) Two long straight parallel wires 10 cm apart carry currents of 5 A and 8 A in the same direction. Calculate the force per unit length between them and state whether it is attractive or repulsive. *(Apply)* [1]

(ii) A galvanometer of resistance 50 Ω gives full-scale deflection for a current of 2 mA. How can it be converted into a voltmeter of range 0–20 V? Find the resistance of the voltmeter. *(Apply)* [2]

Question 13

An optics student uses a compound microscope. The examiner gives instructions and asks questions. State whether the response of the student in each case is correct or incorrect. Give a reason for your answer. *(Evaluate)*

(i) **Examiner:** Increase the magnification produced by the objective lens. [1]
Student: Decrease the distance between the object and the objective, keeping in mind that this distance must remain greater than the focal length of the objective.

(ii) **Examiner:** Increase the resolving power of the instrument. [1]
Student: Illuminate the object with light of a longer wavelength.

(iii) **Examiner:** Which of the lenses of focal lengths $f_1 = 1{\cdot}5$ cm, $f_2 = 80$ cm and $f_3 = 5$ cm would you take as the objective and the eyepiece? [1]
Student: $f_1 = 1{\cdot}5$ cm as the objective and $f_2 = 80$ cm as the eyepiece.

Question 14

Using Huygens' wave theory and a labelled diagram, show that when a plane wave is refracted from a medium of refractive index n_1 into one of refractive index n_2, $n_1 \sin i = n_2 \sin r$. *(Understand)* [3]

Question 15

In Young's double-slit experiment with monochromatic light, the screen is moved $6{\cdot}0 \times 10^{-2}$ m towards the slits and the fringe width changes by $3{\cdot}0 \times 10^{-5}$ m. If the distance between the slits is $1{\cdot}2 \times 10^{-3}$ m, calculate the wavelength of the light used. *(Apply)* [3]

Question 16

(i) Draw a neat labelled ray diagram of the image formed by an astronomical telescope when the final image is at the least distance of distinct vision, and state the formula for its angular magnification. *(Understand)* [2]

(ii) An astronomical telescope has an objective of focal length 150 cm and an eyepiece of focal length 5 cm. Calculate its magnifying power when the final image is formed at 25 cm from the eye. *(Apply)* [1]

Question 17

(i) Deduce the expression for the de Broglie wavelength of an electron accelerated from rest through a potential difference V. *(Understand)* [2]

(ii) An electron beam has a kinetic energy of 64 eV. Find the wavelength associated with the beam. *(Apply)* [1]

SECTION D – 15 MARKS

Question 18

(i)(a) A resistor R, an inductor L and a capacitor C are connected in series to an a.c. source. Using a phasor diagram, derive an expression for the impedance of the circuit. *(Understand)* [3]

(b) A galvanometer has 40 divisions on its scale and gives a deflection of 10 divisions per mA. Its resistance is 50 Ω. A shunt of 2 Ω is connected across it. What is the maximum current that the galvanometer can now read? *(Apply)* [2]

OR

(ii)(a) A conducting rod PQ of length l is pulled with a constant velocity v over two parallel rails joined by a resistor R, in a uniform magnetic field B perpendicular to the plane of the rails. Deduce expressions for (1) the induced current, (2) the force necessary to pull the rod, and (3) the power delivered by the external source. Draw the necessary diagram. *(Understand)* [3]

(b) What are eddy currents? Write one use of them. *(Recall)* [2]

Question 19

(i)(a) Using Bohr's postulates, derive an expression for the orbital velocity of the electron in the nth orbit of a hydrogen-like atom of atomic number Z. *(Understand)* [3]

(b) Hence show that the speed of the electron in the first orbit of hydrogen is about 1/137 of the speed of light. *(Apply)* [1]

(c) Some energy levels of the hydrogen atom are −13·6 eV, −3·40 eV, −1·51 eV and −0·850 eV. A line of wavelength 658 nm is seen in its emission spectrum. Deduce the energy of the photon, and identify the transition that gives rise to it. *(Apply)* [1]

OR

(ii)(a) Sketch a graph showing the variation of the binding energy per nucleon of a nucleus with its mass number. Mark on it the regions of nuclei that are (1) stable, (2) prone to fission, (3) prone to fusion. *(Understand)* [3]

(b) Calculate the energy released in the nuclear reaction $^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He}$ + energy. (Mass of $^{2}_{1}\mathrm{H}$ = 2·014102 u, mass of $^{4}_{2}\mathrm{He}$ = 4·002604 u, 1 u = 931·5 MeV/c^2) *(Apply)* [2]

Question 20

A rectifier is an electronic circuit that converts alternating current into direct current. Ravi builds one for his school laboratory. He starts with a p–n junction diode and studies how the depletion region changes with the bias. A half-wave rectifier uses a single diode, so only one half of every cycle reaches the load and its output is a poor, pulsating d.c. Ravi therefore builds a full-wave rectifier from two diodes and a centre-tapped transformer, powered by the 50 Hz mains, and compares the two outputs. He also plots the forward and reverse characteristics of the junction diode, and finally uses a Zener diode to keep the output voltage constant.

(i) State how the thickness of the depletion region of a p–n junction changes in forward bias and in reverse bias. *(Recall)* [1]

(ii) Draw a neat, labelled circuit diagram of the full-wave rectifier built by Ravi. *(Understand)* [1]

(iii) Show the input and the output waveforms for this rectifier. *(Understand)* [1]

(iv) Sketch the forward and the reverse characteristic curves of the junction diode, and mark the knee (threshold) voltage. *(Understand)* [1]

(v) A Zener diode of breakdown voltage 6·0 V is connected in the reverse direction across the output through a 300 Ω series resistor from a 12 V supply (no load connected). Find the current through the series resistor. *(Apply)* [1]

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